

Huygens' Workshop

toy piano and electronics

Neil Thornock

Program Note

Christiaan Huygens (1629-1695) was an important Dutch mathematician, astronomer, and physicist. Among his many accomplishments, he discovered Saturn's moon Titan and gave the first accurate description of the rings of Saturn. To improve astronomical observations, he also developed the first accurate pendulum clock (achieving accuracy to one minute per day) and later constructed the first accurate spring-regulated clock. He is also credited with one of the earliest theses discussing extraterrestrial life and with a treatise describing 31-tone equal temperament.

I tried to touch on each of these aspects of his career in the two movements of this work. The title *Saturn's Child* refers both to its moon and to the fictive possibility of alien life on Saturn, while the microtonal gestures employ 31-tone equal temperament in a rather cursory nod to his treatise. The music depicts a space explorer orbiting around Saturn, passing through one of its rings (the "stardust" material), coming near collision with Titan, and finally engaging in an austere, transcendent meeting with life from space.

The second movement depicts a clock shop – perhaps an old Renaissance cottage shop in which all the clocks are initially set for the same time, gradually going out of sync, creating pandemonium, and leading to the explosion and demolition of the old cottage – the sole survivor being a trusty grandfather clock, faithfully beating away its inaccurate version of time. The toy piano seemed a perfect instrument for this role: the tinkling sound is reminiscent of dusty, cheap clock bars, and the inaccurate playing mechanism parallels the (by today's standards) inaccurate timekeeping of days gone by.

Performance Notes

Performance requires an amplified toy piano of 1-1/2 octave range (C to F) and Max/MSP. Instructions pertaining to the electronics are included in the main Max patch window.

An audio CD intended for practice purposes will be provided upon request. The audio CD contains the prerecorded material for both movements as well as a realization of the complete piece.

Notes on *Saturn's Child*: Rhythmic notation (except in the dance section) is approximately spatial. Rhythmic freedom is desired. The alignment of toy piano to electronics is also somewhat flexible but should generally align as notated in the score. The recorded realization on the CD is meant only as a guide, not as a definitive interpretation.

Note on *Clockwise*: Kick it!!

Movements may be performed separately if desired.

Duration: ca. 10 min.

Please contact the composer regarding any performances:
neilthornock@gmail.com

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I. Saturn's Child

Toy Piano

Hit RETURN key to begin

CD track

0:26

low big hit

2

0:47

0:50

supple, not fast

T. Pno.

reverse attacks

p

microtonal weirdvibes

CD

3

0:59

1:12

1:16

f

junk metal

mf

T. Pno.

CD

4

1:27

1:32

mp

T. Pno.

CD

5 accidentals apply to repeated notes throughout the mvt.

T. Pno.

mf

6

T. Pno.

1:49 1:53 2:06

f *p* *p*

CD

microtonal chord

7 Dance it! ♩ = ca. 137.14

T. Pno.

mf

CD

junk metal dance

11 Faster ♩ = 160

T. Pno.

CD

15

T. Pno.

CD

19

T. Pno.

CD

23 **Faster!** ♩ = 192

T. Pno.

f

CD

add weirdvibe dance

27

T. Pno.

CD

31 **Wild presto!!** ♩ = 240

T. Pno.

ff

CD

35

T. Pno.

CD

39

T. Pno.

CD

2:50

42

T. Pno.

CD

background sustain gets louder

2:56

43

T. Pno.

CD

Violently!

3:00

fff

* fully chromatic clusters in the range indicated

44

3:05

T. Pno.

CD

45

3:21

3:41

Slowly, peacefully

p

T. Pno.

CD

In the following passage, play each gesture only after the previous note in the electronics becomes audible.

46

4:03

T. Pno.

CD

47

4:28

4:43

5:16

pp

attacca mvmt. 2

T. Pno.

CD

II. Clockwise

Fast, as metronomic as possible ♩ = 120

4 x

7 x

f

3

9

4 x

13

16

T. Pno.

CD

The musical score is divided into five systems, each with a Treble Piano (T. Pno.) and a Cello/Double Bass (CD) staff. The first system includes a tempo instruction 'Fast, as metronomic as possible ♩ = 120' and a dynamic marking '*f*'. It features a 4-measure repeat in the T. Pno. staff and a 7-measure repeat in the CD staff. The second system starts at measure 3 and continues with melodic lines in both staves. The third system starts at measure 9 and includes another 4-measure repeat in the T. Pno. staff. The fourth system starts at measure 13 and the fifth at measure 16, both continuing the melodic development. The score uses various time signatures including 3/8, 4/4, and 3/4, and includes many slurs and articulation marks.

19

T. Pno.

CD

21

T. Pno.

CD

23

T. Pno.

CD

25

T. Pno.

CD

29

T. Pno.

CD

33

T. Pno.

CD

37

T. Pno.

CD

41

T. Pno.

CD

46

T. Pno.

CD

50

T. Pno.

CD

54

T. Pno.

CD

58

T. Pno.

CD

(2 x)

61

T. Pno.

CD

(3 x)

63

T. Pno.

CD

(2 x)

65

T. Pno.

CD

(2 x)

67

T. Pno.

CD

2 x

70

T. Pno.

CD

spectral glissandi

75

T. Pno.

CD

81

T. Pno.

CD

87

T. Pno.

CD

Measures 87-92. The T. Pno. part features a complex melodic line with many accidentals and slurs. The CD part consists of a simple bass line with dotted half notes.

93

T. Pno.

CD

Measures 93-96. The T. Pno. part has a melodic line with slurs and accidentals. The CD part has a bass line with a 3+2 time signature change indicated above the staff.

97

T. Pno.

CD

Measures 97-100. The T. Pno. part continues with a complex melodic line. The CD part has a bass line with dotted half notes and slurs.

100

T. Pno.

CD

Measures 100-103. The T. Pno. part features a melodic line with a 3/4 time signature change indicated above the staff. The CD part has a bass line with dotted half notes.

102

T. Pno.

CD

104

T. Pno.

CD

106

T. Pno.

CD

109

T. Pno.

CD

112

3 x

T. Pno.

CD

114

T. Pno.

CD

116

4 x

8 x

sempre rit.

T. Pno.

CD

8^{vb}